1 Child Mortality in Côte d'Ivoire

1.a I can use the pweight column to balance the class distribution of the sample to match the population distribution. The `pweight` column contains sample weights that modify the contribution of each observation to the overall statistics. It works as a frequency multiplier, where each observation is counted as if it were `pweight` number of observations. In practice, R can use the pweight column to compute weighted statistics, such as means and proportions by using the `weights` argument in functions like `lm()` or `survey::svydesign()`. Internally, weighted functions such as weighted means and lm will multiply each observation by its corresponding weight. For example a weighted mean will be computed using equation Equation 1, and weighted linear regression will minimize the weighted sum of squared residuals, as shown in equation Equation 2.

$$\bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i} \tag{1}$$

$$\sum_{i=1}^{n} w_i (y_i - \hat{y}_i)^2 \tag{2}$$

1.2.a The preferable method is method 2 (M2) because method 1 (M1) underestimates the mortality rate by not capturing the deaths that occur after the survey period. Figure 1 shows graphically the surveyed groups: the x-axis represents the age of the child at the time of the survey, and the y-axis represents the age of the child at the time of death. The red arrow at 45 degrees indicates the frontier of possible outcomes, since by definition no one can die after their current age, the survey captures outcomes below the frontier. Notice M1 captures area B and drops area C underestimating the true mortality rate. M2 captures area C which represents the true mortality rate. For example a surveyed mother has a child one year before the survey and the child dies the year after the survey (area C in Figure 1), M1 will not capture his death because by definition nothing can die after its current age.

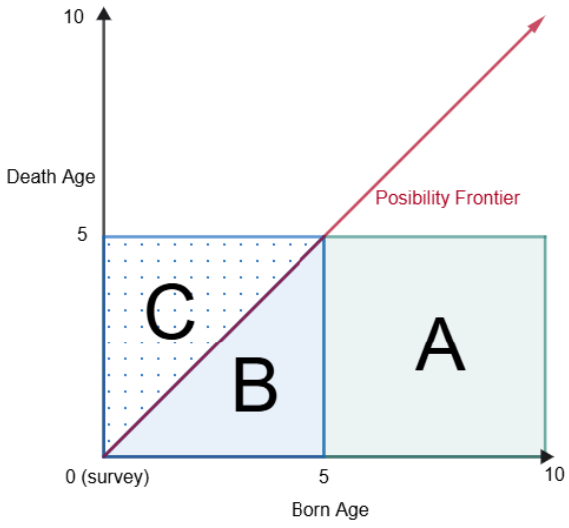
M1

- Pros: it is easy to compute, captures the most recent deaths (recency).
- Cons: underestimates the mortality rate, downward biased, does not capture deaths after the survey period.

M2

- Pros: provides a robust estimate of the true mortality rate.
- Cons: it requires a longer period of time to capture deaths, less recent than M1.

Figure 1: Mortality estimator analysis



1.2.b.i I estimated the mortality rates for boys and girls using both methods in R using the columns `_l5` for method 1 and the difference of total minus `_l5` for method 2 (See appendix for code). The estimated mortality rates for boys and girls using both methods are shown in Table 1.

Table 1: Child mortality rates in Côte d'Ivoire, by sex and method.

	M1 (Boys)	M2 (Boys)	Diff (Boys)	M1 (Girls)	M2 (Girls)	Diff (Girls)	Diff M2 G-B
Mortality rate	0.117 ***	0.130 ***	0.013	0.084 ***	0.124 ***	0.040 ***	-0.006
	(0.007)	(0.008)	(0.011)	(0.007)	(0.008)	(0.010)	(0.012)
N	1431	1332		1428	1291		
R2	0.000	0.000		0.000	0.000		

Significance: *** p < 0.01; ** p < 0.05; * p < 0.1

1.2.b.ii Table 1 shows the estimated child mortality rates in Côte d'Ivoire for boys and girls using both methods and their differences. The results confirm our theory of M1 underestimating the true mortality rate. We can clearly see the method estimates differences are positive indicating that M1 systematically underestimates the true mortality rate estimated by M2. Using the best method (M2), the average mortality rate for boys is 13.5% and for girls is 11.9%. The difference between girls and boys is -2 percentage points and statistically significant at the 90% confidence level, indicating that boys have a higher mortality rate than girls.

2 Health status, nutritional status and wealth in Côte d'Ivoire

2.1.a Let's denote the estimated mean weight as W_{obs}, the true mean weight as W_{true}, and the proportion of weights measured in hectograms as p_h. The observed mean weight can be expressed as the sum of the mean weights measured in kg and in hectograms multiplied by their corresponding probabilities:

$$\begin{aligned} E[W_{\text{obs}}] &= (1 - p_h) \cdot W_{\text{true}} + p_h \cdot W_{\text{true}} \cdot 10 \\ &= W_{\text{true}} \cdot (1 - p_h + 10p_h) \\ &= W_{\text{true}} \cdot (1 + 9p_h) \end{aligned}$$

Thus, having an estimate of p_h would allow us to correct the observed mean weight. Notice that if $p_h = 0$, then $E[W_{\text{obs}}] = W_{\text{true}}$, and if $p_h = 1$, then $E[W_{\text{obs}}] = 10W_{\text{true}}$.

For applying the correction, it is necessary to have a good estimate of p_h. In this case, it can be approximated by the proportion of weights above 25kg in 2008 which is 30.4% of the observations. Using this, we can correct the mean weight estimate by dividing the observed mean weight by $(1 + 9p_h)$. See Table 2 where the 2008 mean weight is corrected. After correction, the mean weight in 2008 is 11.48kg, which is consistent with previous years.

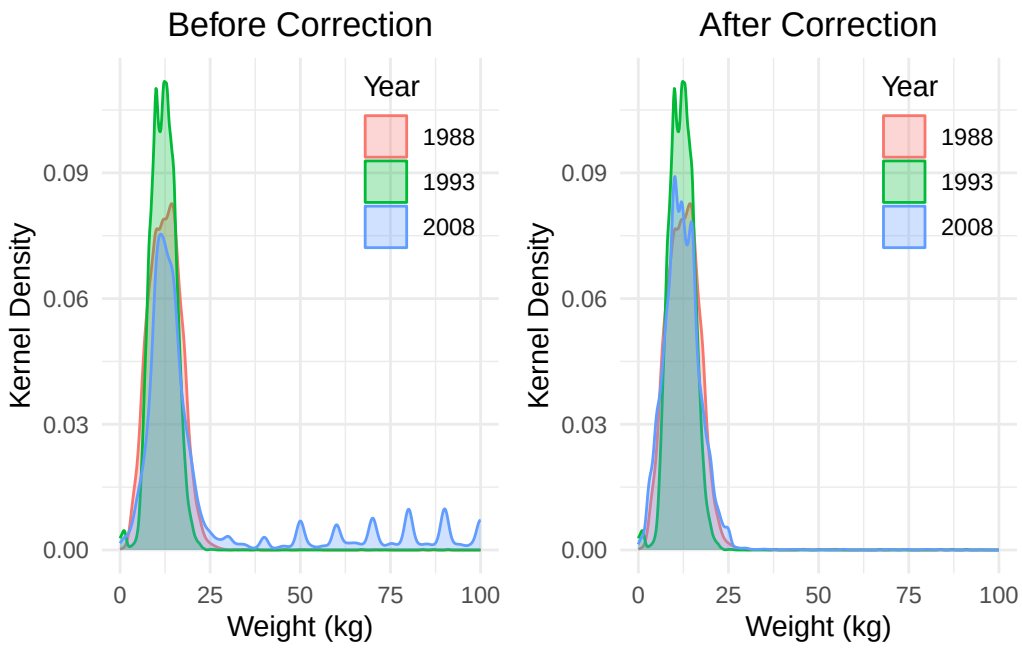
Table 2: Summary of Weight Estimates With Hectogram Correction

Year	Obs	Mean Weight	Mean Weight Corrected
1988	6475	12.46	12.46
1993	7109	11.94	11.94
2008	7333	42.87	11.48

2.1.b By looking at the original distribution the weight measurements in 2008 show a wider distribution than the 1988 and 1993 distributions. The 2008 distribution shows a principal component at around 12kg but also shows multiple peaks at weight > 25kg while the 1988 and 1993 distributions show a principal component at around 12kg with minimal outliers e.g. a few values below 1kg. See Figure 2 LHS.

After updating the 2008 weight distribution by dividing all weights > 25kg by 10, the distribution shows no peaks after 25kg. See Figure 2 RHS.

Figure 2: Distribution of Weight (kg) Before and After Correction



2.2 Estimated z-scores are computed using the mentioned formula i.e. (height - medheight) / stdheight in R.

2.3.a As shown in Table 3, the national stunting rate for 2-4 years old children with outliers is 24.1% in 1988, 42% in 1993, and 46.4% in 2008. This increasing trend shows an overall increment of 22.3 percentage points in the stunting rate over the 20-year period.

The north and south regions show an increasing trend over time, but the north region shows a higher stunting rate than the south region in 1988 and 2008. Particularly, in 1988 the difference between north and south is 5.6 percentage points at the 1% significance level, and in 2008 the difference is 4.5 percentage points at the 5% significance level. In 1993 the difference is not statistically significant different from zero, indicating that the north and south regions had similar stunting rates in 1993. Many theories can explain the differences between north and south e.g. differences in measurement, differences in the population, or the effect of socioeconomic factors such as poverty-alleviating programs, and access to health care.

The results excluding outliers show a similar trend in 1988 and 1993, however, in 2008 the stunting is not as different from 1993 except in the north region where there is a difference of 4 percentage points. This change in the pattern suggest that the outliers correction had an impact on the results and drafted conclusions. Regional differences between north and south persist, there is still a significant difference of 6.1 and 3.5 percentage points in 1988 and 2008 at the 1% and 10% significance levels respectively between the north and south regions despite the change in the stunting rate over time pattern.

2.3.b As shown in Table 4, the stunting rate for boys and girls is increasing over time for the model with outliers (LHS four models). Starting from 1988, boys and girls had a stunting rate of 24.6% and 23.6% with no significant difference. In 1993, boys and girls had a stunting rate of 43% and 40% with a significant difference of 3 percentage points indicating boys had a higher stunting rate. In 2008, boys and girls had a stunting rate of 47% and 45% respectively with no significant difference.

The stunting rates pattern without outliers (RHS four columns) changes in 2008. In this period, removing the outliers lowered the national stunting rates from 47% with outliers to 41% without outliers. Differences between boys and girls are significant at 5% significance level in 1993, while there is no significant difference in 1988 and 2008.

2.4 Figure 3a shows the relationship between log household consumption per capita (binned into ventiles) and the probability of being stunted for 2-4 year-olds. The vertical shift of the fitted regression lines shows a clear trend of increasing stunting rates across years. The negative slopes indicate higher consumption lowers stunting probability.

Table 3: National, North, and South stunting rates for 2-4 years old children excluding outliers.

	National	North	South	Diff (North-South)	National	North	South	Diff (North-South)
Year 1988	0.241 ***	0.282 ***	0.226 ***	0.056 ***	0.239 ***	0.284 ***	0.222 ***	0.061 ***
	(0.009)	(0.016)	(0.011)	(0.020)	(0.009)	(0.016)	(0.010)	(0.020)
Year 1993	0.420 ***	0.409 ***	0.423 ***	-0.014	0.412 ***	0.397 ***	0.416 ***	-0.020
	(0.007)	(0.014)	(0.008)	(0.017)	(0.007)	(0.014)	(0.008)	(0.017)
Year 2008	0.464 ***	0.498 ***	0.453 ***	0.045 **	0.410 ***	0.436 ***	0.402 ***	0.035 *
	(0.008)	(0.015)	(0.009)	(0.018)	(0.008)	(0.016)	(0.010)	(0.019)
Outliers	Yes	Yes	Yes	Yes	No	No	No	No
N	11755	3194	8561		10897	2930	7967	
R2	0.410	0.421	0.407		0.381	0.384	0.381	

Significance: *** p < 0.01; ** p < 0.05; * p < 0.1

Table 4: National, Male, and Female stunting rates for 2-4 years old children.

	National	Male	Female	Diff (Male-Female)	National	Male	Female	Diff (Male-Female)
Year 1988	0.241 ***	0.246 ***	0.236 ***	0.011	0.239 ***	0.245 ***	0.233 ***	0.011
	(0.009)	(0.012)	(0.013)	(0.018)	(0.009)	(0.012)	(0.013)	(0.018)
Year 1993	0.420 ***	0.434 ***	0.405 ***	0.029 **	0.412 ***	0.425 ***	0.398 ***	0.028 **
	(0.007)	(0.010)	(0.010)	(0.014)	(0.007)	(0.010)	(0.010)	(0.014)
Year 2008	0.464 ***	0.473 ***	0.453 ***	0.020	0.410 ***	0.420 ***	0.398 ***	0.022
	(0.008)	(0.010)	(0.011)	(0.015)	(0.008)	(0.011)	(0.012)	(0.016)
Outliers	Yes	Yes	Yes	Yes	No	No	No	No
N	11755	6072	5683		10897	5610	5287	
R2	0.410	0.421	0.398		0.381	0.392	0.369	

Significance: *** p < 0.01; ** p < 0.05; * p < 0.1

The 1993 regression line is notably steeper than in 1988 and 2008, indicating a more pronounced socioeconomic gradient that year.

Analytically, the results can be confirmed through the linear regression model in Equation 3. Where $log(\widetilde{\text{Conspc88}}_i)$ is the log of household consumption per capita centered around its mean, β_0 is the stunting rate for the mean child household consumption, and β_1 is the effect of log household consumption on the probability of being stunted in 1988. β_2 and β_3 capture the change in the stunting rate in 1993 and 2008 relative to 1988, while β_4 and β_5 capture the change in the effect of log household consumption on the probability of being stunted in 1993 and 2008 relative to 1988. Z_i is a vector of control variables including sex, age, and education of the household head. The results for all population and for each tercile are summarized in Table 5.

$$\begin{aligned} Stunted_i = \beta_0 + \beta_1 log(\widetilde{\text{Conspc88}}_i) + \beta_2 Y93_i + \beta_3 Y08_i + \\ (\beta_4 Y93_i \times log(\widetilde{\text{Conspc88}}_i)) + \beta_5 (Y08_i \times log(\widetilde{\text{Conspc88}}_i)) + Z_i + \epsilon_i \end{aligned} \tag{3}$$

$$\text{where } \widetilde{\text{Conspc88}}_i = log(\text{Conspc88}_i) - \overline{log(\text{Conspc88})}$$

According to Table 5, the 1988 baseline stunting rate is 37%, where a 1% consumption increase decreases stunting

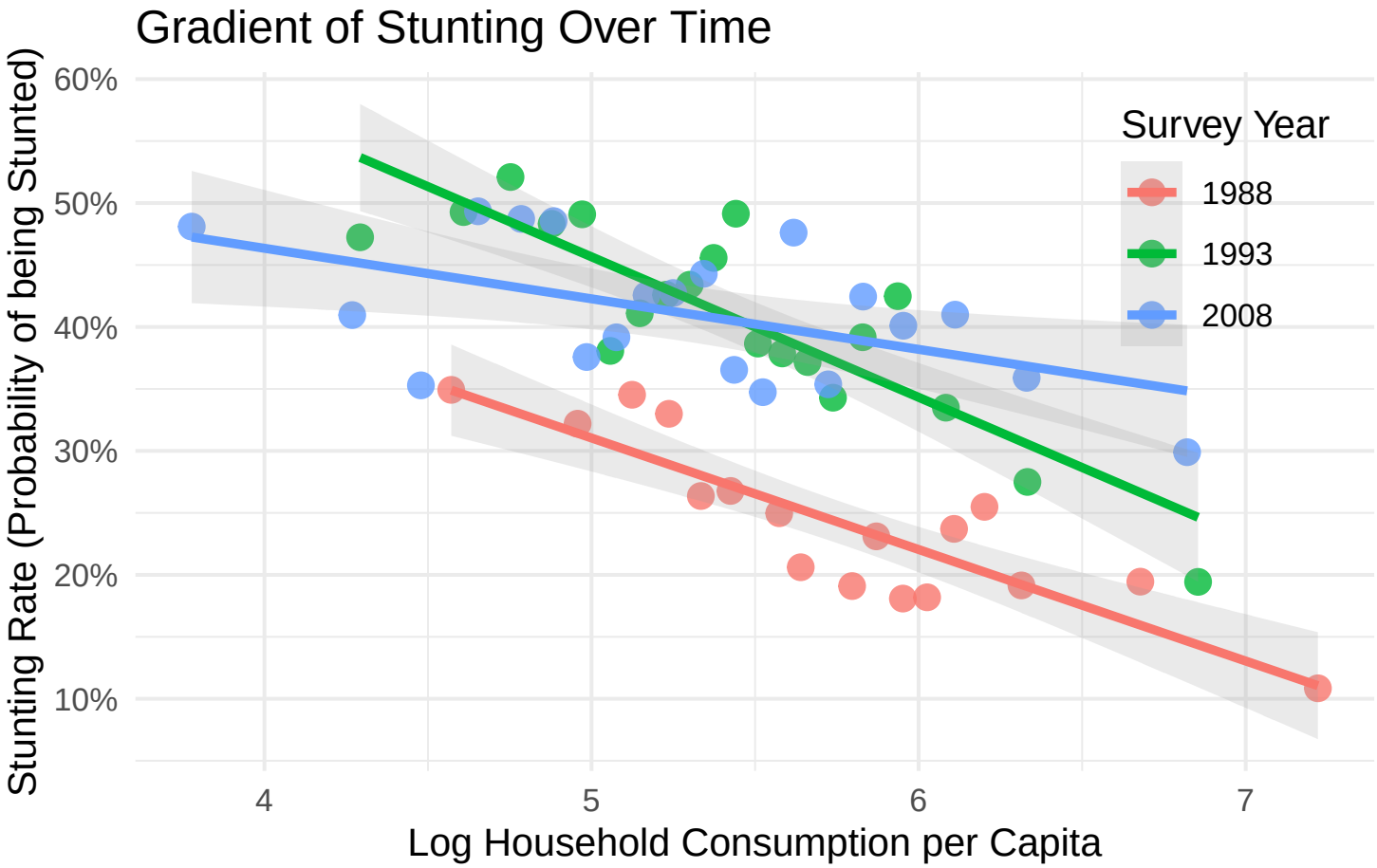
probability by 0.07 percentage points. Baseline stunting shifted 13 and 15 percentage points higher in 1993 and 2008, respectively. The consumption gradient changed slightly in 2008, with a 1% consumption increase associated with a 0.04 percentage point higher penalty relative to 1988.

Analyzing by consumption terciles reveals heterogeneous patterns. In 1988, low-consumption children had a 42% stunting rate versus 33% for high-consumption. By 1993, stunting increased by 15 and 21 percentage points for medium and high-consumption households respectively, while remaining unchanged for low-consumption households. The protective effect of consumption was significantly stronger in 1993 exclusively for the high-consumption tercile, where a 1% consumption increase reduced stunting by an additional 0.12 percentage points relative to 1988 (and 0.08 points in 2008, significant at 10%).

Overall, baseline stunting rates increased over time while the protective effect of household consumption fluctuated, becoming notably steeper in 1993. The stratified model confirms this relationship is not uniform: high-consumption households experienced a significantly more pronounced protective effect of consumption against stunting in 1993 than in 1988 or 2008.

Figure 3: Log household consumption ventiles vs stunting rate for 2-4 years old children by year

(a) Outliers are removed from the analysis. Includes linear regression lines with 95% confidence intervals.



3 A meaningful table (6 pts)

The Tiebout model predicts individuals will "vote with their feet" to relocate for optimal public services and economic opportunities, though mobility remains heavily constrained by structural frictions like relocation costs and border regimes. Here, I evaluate the average share of foreign populations (non-citizens) across OECD countries (2013–2023) to understand how national policies shape demographic stocks.

Table 5: Heterogeneous Effects of Consumption on Stunting Rates by Consumption Tertile

	General (All)	Low Cons (T1)	Medium Cons (T2)	High Cons (T3)
Average Stunting Rate (1988)	0.371 ***	0.421 ***	0.393 ***	0.333 ***
	(0.021)	(0.063)	(0.036)	(0.039)
Log Consumption (Centered)	-0.074 ***	0.026	-0.136	-0.009
	(0.014)	(0.083)	(0.101)	(0.027)
Year 1993 Shift	0.138 ***	0.031	0.156 ***	0.218 ***
	(0.012)	(0.062)	(0.019)	(0.034)
Year 2008 Shift	0.150 ***	0.061	0.158 ***	0.257 ***
	(0.013)	(0.062)	(0.023)	(0.038)
Cons. Effect Diff (1993)	-0.023	-0.124	-0.047	-0.121 ***
	(0.018)	(0.092)	(0.126)	(0.041)
Cons. Effect Diff (2008)	0.047 ***	-0.032	0.020	-0.081 *
	(0.018)	(0.088)	(0.141)	(0.045)
Controls	Yes	Yes	Yes	Yes
Outliers Excluded	Yes	Yes	Yes	Yes
N	10897	3634	3631	3632
R2	0.043	0.009	0.032	0.053

Significance: *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Figure 4a illustrates the OECD average foreign population share, capturing long-term residents (over 12 months) while excluding temporary visitors. This non-citizen share grew from 5.5% in 2013 to 7.2% in 2023, reflecting increasing international mobility. OECD-level confidence intervals are displayed, while individual country-level intervals are omitted due to data aggregation.

Figure 4a demonstrates cross-country variance driven by distinct policy models and historical migration patterns:

1. **Above the mean**: Germany has a high foreign population share (15.8% in 2023), driven by historic labor and humanitarian inflows and active policy engineering designed to attract skilled foreign workers.
2. **Below the mean**: Poland exhibits a low share (1.1% in 2023) due to post-WWII demographic homogeneity, an economic transition that historically exported labor, and immigration policies favoring temporary rather than permanent structural migration.
3. **Near the mean**: The U.S. (8.4%) and France (7.1%) maintain strong international appeal, but robust naturalization pathways continuously convert residents into citizens, removing them from the "foreign population" metric and keeping stocks stable.

Evaluating these differences is critical as nations with high stocks must enhance foreigners integration, whereas countries with suppressed shares must streamline visa processes to attract global talent.

Appendix R Script

Figure 4: Stock of foreign population (non-citizens) over time

(a) The dashed line represents the OECD stock of foreign population (non-citizens), while the solid lines are selected countries.

